

FILTER DESIGN TEST CASE

Duplexer Module - n79 (4.4-5.0 GHz 5G NR) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting n79 (4.4-5.0 GHz 5G NR) for Automotive V2X Communication. The filter is designed on AlN on Si substrate with a target center frequency of 1152.6 MHz and bandwidth of 347.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: n79 (4.4-5.0 GHz 5G NR)
- Application: Automotive V2X Communication
- Substrate: AlN on Si
- Package: CSP (Chip Scale Package)
- Design Tool: PathWave EM Design
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Duplexer Module design targeting n79 (4.4-5.0 GHz 5G NR). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1152.6 MHz
Bandwidth (3 dB):	347.1 MHz
Insertion Loss Target:	≤ 2.9 dB
Return Loss Target:	≥ 15.0 dB
Out-of-Band Rejection:	≥ 39.0 dB
Isolation (Duplexer):	≥ 32.0 dB
Q Factor:	2528
Temperature Coefficient:	-14.5 ppm/°C
Die Size:	2.1 x 1.5 mm
Wafer Size:	6-inch
Number of Resonators:	7
Electrode Material:	Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

- 1. Open PathWave EM Design and load the Duplexer Module template project.
- 2. Define the substrate stack: AlN on Si with specified layer thicknesses.
- 3. Set design targets: center freq 1152.6 MHz, BW 347.1 MHz.
- 4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
- 5. Optimize resonator geometry using gradient descent (target IL < 2.9 dB).
- 6. Verify coupling coefficients between resonator stages.
- 7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
- 8. Export GDS-II layout for mask fabrication.
- 9. Fabricate prototype wafers (6-inch, AlN on Si).
- 10. Perform on-wafer probing using Keysight E5080B ENA.
- 11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3458 MHz.
- 12. Compare measured results against simulation and specification limits.
- 13. Perform temperature cycling (-40°C to +85°C) and re-measure.
- 14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.9 dB
- Return loss in passband shall be >= 15.0 dB
- Out-of-band rejection at +/- 521 MHz offset >= 39.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 2.1 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-06-10
Devices Tested: 74
Insertion Loss (meas): 2.59 dB
Return Loss (meas): 17.4 dB
Rejection (meas): 41.7 dB
Isolation (meas): 36.9 dB
Q Factor (meas): 2528
Group Delay Variation: 1.2 ns
Power Handling: +31 dBm
Die Yield: 97.5%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated excellent performance for n79 (4.4-5.0 GHz 5G NR).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

Author: Mark Sullivan Date: 2024-06-10
Reviewer: Dr. Takeshi Yamamoto Date: 2024-06-16
Approver: Dr. Yuko Ishida Date: 2024-06-15

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